Appendix A. The Number Systems

Natural numbers, integers, rationals, and the construction of the reals by Dedekind cuts

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Chapter 1 took the real numbers as a complete ordered field. Here we construct such a field. We accept the arithmetic and order of $\mathbb{N} = \{0, 1, 2, \dots\}$, including the fact that every nonempty subset of $\mathbb{N}$ has a least element, build $\mathbb{Z}$ and $\mathbb{Q}$ by equivalence relations, and obtain $\mathbb{R}$ from Cauchy sequences of rationals.

A.1 Integers and rationals

Motivation. Chapter 1 took the real numbers as an axiomatic starting point. This appendix shows that such a field actually exists, by building it from the rationals; the first step is to recall how $\mathbb{Z}$ and $\mathbb{Q}$ themselves arise.

On $\mathbb{N} \times \mathbb{N}$ put $(m, n) \sim (m', n') \iff m + n' = n + m'$ (think of $m - n$); this is an equivalence relation, and $(m, n) \sim (m - k, n - k)$ when $m, n \geq k$. $\mathbb{Z}$ is the set of classes $[m, n]$, with $[m, n] \geq [k, l] \iff m + l \geq n + k$, $[m, n] + [k, l] = [m + k, n + l]$, $[m, n] \cdot [k, l] = [mk + nl, ml + nk]$. These are well defined (independent of representatives), and every integer is $[n, 0]$, $[0, 0]$ or $[0, n]$ for a unique $n \geq 1$, written n, 0, $-n$; the map $n \mapsto [n, 0]$ embeds $\mathbb{N}$ preserving operations and order.

Theorem A.1 (Theorem 1)

(a) For $a, b \in \mathbb{Z}$, $a \geq b$ or $b \geq a$.
(b) A nonempty subset of $\mathbb{Z}$ bounded above has a largest element, and one bounded below has a least element.

Proof. (a) reduces to the same property of $\mathbb{N}$ for $m + l$ and $n + k$.

(b) If S is bounded below by b, then $\{s - b : s \in S\} \subset \mathbb{N}$ has a least element k, and $b + k$ is least in S; for S bounded above apply this to $-S$. ∎

$\mathbb{Z}$ satisfies the field axioms except the existence of multiplicative inverses, and with $P_{\mathbb{Z}} = \{1, 2, \dots\}$ the order axioms. On $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ put $(a, b) \sim (c, d) \iff ad = bc$ and order the classes by $(a, b) > (c, d) \iff abd^2 > cdb^2$; the classes are the rationals a/b, with $a/b + c/d = (ad + bc)/bd$ and $(a/b)(c/d) = ac/bd$. One checks that $\mathbb{Q}$ is an ordered field containing $\mathbb{Z}$, so the results of Section 1.1, valid in any ordered field, hold in $\mathbb{Q}$.

A.2 Real numbers as classes of Cauchy sequences

Motivation. A real number is what a Cauchy sequence of rationals "should" converge to. Taking equivalence classes of such sequences makes this literal and produces a complete ordered field, which justifies the axioms of Chapter 1.

A rational sequence $(a(i))$ is a **fundamental sequence** if for every rational $\epsilon > 0$ there is N with $|a(i) - a(j)| < \epsilon$ for $i, j \geq N$.

Theorem A.2 (Theorem 2)

(a) Fundamental sequences are bounded.
(b) A bounded monotone rational sequence is fundamental.
(c) If $(a(i))$ is fundamental, so is $(|a(i)|)$.
(d) If $(a(i)), (b(i))$ are fundamental, so are $(a(i) \pm b(i))$ and $(a(i)b(i))$.
(e) If $(a(i))$ is fundamental and there are N and a rational $d > 0$ with $|a(i)| \geq d$ for $i \geq N$, then $(1/a(i))_{i \geq N}$ is fundamental.

Proof. (a) as in Proposition 2.12.

(b) If an increasing sequence bounded by B were not fundamental, there would be a rational $\epsilon > 0$ and indices $i_1 < j_1 \le i_2 < j_2 \le \cdots$ with $a(j_k) - a(i_k) \ge \epsilon$; then $a(j_k) \ge a(i_1) + k\epsilon > B$ for large k.

(c) $||a(i)| - |a(j)|| \le |a(i) - a(j)|$.

(d) $|a(i)b(i) - a(j)b(j)| \le |a(i)||b(i) - b(j)| + |b(j)||a(i) - a(j)|$ and (a).

(e) $|1/a(i) - 1/a(j)| \le |a(i) - a(j)|/d^2$. ■

Two fundamental sequences are **equivalent**, $(a(i)) \sim (b(i))$, if for every rational $\epsilon > 0$ there is N with $|a(i) - b(i)| < \epsilon$ for $i \ge N$; this is an equivalence relation. A **real number** is an equivalence class $[a]$; $\mathbb{R}$ is the set of classes. Define $[a] > [b]$ if there are a rational $d > 0$ and N with $a(i) - b(i) > d$ for $i \ge N$; this does not depend on the representatives.

Theorem A.3 (Theorem 3, trichotomy)

For $[a], [b] \in \mathbb{R}$ exactly one of $[a] > [b]$, $[a] = [b]$, $[b] > [a]$ holds.

Proof. For rational $\epsilon > 0$ let N_ϵ be such that $|a(i) - a(N_\epsilon)| < \epsilon$ and $|b(i) - b(N_\epsilon)| < \epsilon$ for $i \ge N_\epsilon$. Then for $i \ge N_\epsilon$,

$$d_\epsilon := a(N_\epsilon) - b(N_\epsilon) - 2\epsilon < a(i) - b(i) < a(N_\epsilon) - b(N_\epsilon) + 2\epsilon =: d'_\epsilon.$$

If $d_\epsilon > 0$ for some ϵ, then $[a] > [b]$; if $d'_\epsilon < 0$ for some ϵ, then $[b] > [a]$. Otherwise for every ϵ, $|a(N_\epsilon) - b(N_\epsilon)| \le 2\epsilon$, so $|a(i) - b(i)| < 4\epsilon$ for $i \ge N_\epsilon$, and $[a] = [b]$. Two of the cases cannot hold together, by the definitions. ■

Define $[a]+[b] = [a+b]$ and $[a][b] = [ab]$ (termwise operations); these are well defined by Theorem A.2. For $r \in \mathbb{Q}$ let r^* be the class of the constant sequence r; $r \mapsto r^*$ embeds $\mathbb{Q}$ in $\mathbb{R}$ preserving operations and order. 0^* and 1^* are the identities, $[-a]$ is the additive inverse, and if $[a] \ne 0^*$ then by Theorem A.3 there are $d > 0$ and N with $|a(i)| \ge d$ for $i \ge N$, so the class of $(1/a(i))_{i \ge N}$ (Theorem A.2(e)) is the inverse. With $P = \{[a] : [a] > 0^*\}$, $\mathbb{R}$ is an ordered field, and $|[a]| = [|a|]$.

Theorem A.4 (Theorem 4, density of $\mathbb{Q}$)

If $[b] > [a]$, there is a rational r with $[b] > r^* > [a]$.

Proof. Choose N_1 and rational $d > 0$ with $b(i) - a(i) > d$ for $i \ge N_1$, and N_2 with $|a(i) - a(j)| < d/4$, $|b(i) - b(j)| < d/4$ for $i, j \ge N_2$; let $N = \max(N_1, N_2)$ and $r = a(N) + d/2$. For $i \ge N$, $r - a(i) = d/2 + (a(N) - a(i)) > d/4$, so $r^* > [a]$; and $b(i) - r = (b(i) - b(N)) + (b(N) - a(N)) - d/2 > -d/4 + d - d/2 = d/4$, so $[b] > r^*$. ■

Theorem A.5 (Theorem 5, completeness)

Every Cauchy sequence $([a_n])$ in $\mathbb{R}$ converges in $\mathbb{R}$.

Proof. If the sequence is eventually constant there is nothing to prove; otherwise, passing to a subsequence (which suffices by Lemma 2.13, whose proof is valid in any ordered field), assume $[a_n] \ne [a_{n+1}]$ for all n. By Theorem A.4 choose rationals $r(n)$ strictly between $[a_n]$ and $[a_{n+1}]$; then $|[a_n] - r(n)^*| < |[a_n] - [a_{n+1}]| \to 0$. The rational sequence $(r(n))$ is fundamental, since $|r(n) - r(m)|^* \le |r(n)^* - [a_n]| + |[a_n] - [a_m]| + |[a_m] - r(m)^*|$. Let $[r]$ be its class. Given $[e] > 0$ choose, by Theorem A.4, a rational $\epsilon > 0$ with $\epsilon^* < [e]$, and M with $|r(i) - r(j)| < \epsilon/2$ for $i, j \ge M$. For $n \ge M$, $|r(n)^* - [r]| \le (\epsilon/2)^*$, because the termwise differences $r(n) - r(i)$ are eventually below $\epsilon/2$. Hence for large n, $|[a_n] - [r]| \le |[a_n] - r(n)^*| + (\epsilon/2)^* < [e]$. ■

Finally, Theorems A.4 and A.5 imply the completeness axiom: given S nonempty and bounded above, the bisection argument after Theorem 1.20 produces nested intervals with rational end points

whose lengths tend to 0; the left end points form a Cauchy sequence, whose limit (Theorem A.5) is $\sup S$. So $\mathbb{R}$ is a complete ordered field, unique up to isomorphism by Theorem 1.10.